

AWS

1) Compute Services

- **Amazon EC2 – Virtual servers:** Provides resizable virtual machines in the cloud to run any application.
 - **EC2 Auto Scaling:** Automatically adds or removes EC2 instances based on traffic or load.
 - **AWS Lambda – Serverless compute:** Runs code without managing servers and scales automatically.
 - **Amazon Lightsail:** Simple, low-cost VPS for small apps and websites.
 - **AWS Elastic Beanstalk:** Deploy and manage apps without handling infrastructure.
 - **AWS Batch:** Runs large-scale batch and high-performance computing jobs.
 - **AWS App Runner:** Quickly deploy web apps from source or containers.
 - **AWS Outposts:** Brings AWS services to on-premise environments.
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2) Storage Services

- **Amazon S3 – Object storage:** Stores and retrieves any amount of data securely.
 - **Amazon S3 Glacier – Archival storage:** Low-cost storage for long-term backups and archives.
 - **Amazon EBS – Block storage:** Persistent disk storage for EC2.
 - **Amazon EFS – File storage:** Shared file system for multiple EC2 instances.
 - **AWS Backup:** Centralized backup service for AWS resources.
 - **AWS Storage Gateway:** Connects on-premise storage to AWS cloud.
 - **AWS Snow Family:** Physical devices to transfer large data to AWS.
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3) Database Services

- **Amazon RDS:** Managed relational databases with automatic maintenance.
- **Amazon Aurora:** High-performance cloud-native relational database.
- **Amazon DynamoDB:** Serverless NoSQL key-value database.
- **Amazon Redshift:** Data warehouse for analytics at scale.
- **Amazon ElastiCache:** In-memory cache using Redis or Memcached.
- **Amazon Neptune:** Graph database for relationships.
- **Amazon Keyspaces:** Managed Cassandra-compatible database.

- **Amazon Timestream:** Time-series database for IoT and metrics.
 - **Amazon DocumentDB:** MongoDB-compatible document database.
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4) Networking & Content Delivery

- **Amazon VPC:** Private virtual network in AWS.
 - **Elastic Load Balancing (ELB):** Distributes traffic across servers.
 - **Amazon Route 53:** DNS and domain management service.
 - **Amazon CloudFront:** Global content delivery network.
 - **AWS Direct Connect:** Dedicated private connection to AWS.
 - **AWS Global Accelerator:** Improves global app performance.
 - **AWS App Mesh:** Manages microservices communication.
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5) Security, Identity & Compliance

- **AWS IAM:** Manages users and permissions.
 - **AWS Organizations:** Manages multiple AWS accounts.
 - **AWS KMS:** Manages encryption keys.
 - **AWS CloudHSM:** Hardware-based key storage.
 - **AWS Shield:** DDoS protection.
 - **AWS WAF:** Web application firewall.
 - **Amazon GuardDuty:** Threat detection service.
 - **Amazon Inspector:** Security vulnerability scanning.
 - **AWS Secrets Manager:** Stores and rotates secrets.
 - **AWS Certificate Manager:** Manages SSL/TLS certificates.
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6) Management & Governance

- **Amazon CloudWatch:** Monitors metrics and logs.
- **AWS CloudTrail:** Tracks API activity.
- **AWS Config:** Tracks configuration changes.
- **AWS Systems Manager:** Manages AWS resources centrally.
- **AWS Control Tower:** Sets up multi-account environments.
- **AWS Trusted Advisor:** Best-practice recommendations.

- **AWS Service Catalog:** Manages approved services.
 - **AWS Well-Architected Tool:** Reviews architecture quality.
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7) Analytics & Big Data

- **Amazon Athena:** Query S3 using SQL.
 - **Amazon EMR:** Big data processing with Hadoop/Spark.
 - **AWS Glue:** ETL and data catalog service.
 - **Amazon Kinesis:** Real-time data streaming.
 - **Amazon OpenSearch Service:** Search and log analytics.
 - **Amazon QuickSight:** Business intelligence dashboards.
 - **AWS Data Pipeline:** Data workflow automation.
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11) AI, ML & Generative AI

- **Amazon SageMaker:** Build and deploy ML models.
 - **Amazon Bedrock:** Access foundation models for GenAI.
 - **Amazon Rekognition:** Image and video analysis.
 - **Amazon Comprehend:** Text analysis with NLP.
 - **Amazon Lex:** Chatbot and voice bots.
 - **Amazon Polly:** Text to speech.
 - **Amazon Textract:** Extract text from documents.
 - **Amazon Translate:** Language translation.
 - **Amazon Forecast:** Time-series forecasting.
 - **Amazon Personalize:** Recommendation engine.
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12) IoT Services

- **AWS IoT Core:** Connects and manages IoT devices.
 - **AWS IoT Greengrass:** Run logic on edge devices.
 - **AWS IoT Analytics:** Analyze IoT data.
 - **AWS IoT Device Defender:** Secures IoT devices.
 - **AWS IoT TwinMaker:** Build digital twins.
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Highlighted ones:

Introduction to AWS Cloud Services

AWS provides on-demand computing, storage, networking, and many other services over the internet. You pay only for what you use and can scale resources up or down easily.

Amazon EC2 – Virtual Servers

EC2 lets you run virtual machines in the cloud with full control over OS and software. You can choose instance size, region, and scale as needed.

Amazon S3 – Object Storage

S3 stores files like images, videos, backups, and logs securely. It is highly durable, scalable, and accessible from anywhere.

Amazon S3 Glacier – Archival Storage

Glacier is designed for long-term, low-cost data storage. Data retrieval is slower but much cheaper than regular storage.

Amazon RDS – Managed Databases

RDS runs relational databases without manual setup or maintenance. It handles backups, updates, scaling, and high availability.

Amazon VPC – Private Networking

VPC lets you create a private network in AWS. You control IP ranges, subnets, firewalls, and routing.

Elastic Load Balancing (ELB)

ELB spreads incoming traffic across multiple servers. This improves availability, performance, and fault tolerance.

Amazon Route 53 – DNS Service

Route 53 converts domain names into IP addresses. It also supports health checks and traffic routing.

Amazon CloudFront – Content Delivery

CloudFront delivers content through global edge locations. It reduces latency and speeds up websites and apps.

AWS IAM – Identity & Access

IAM controls who can access AWS resources. You define users, roles, and permissions securely.

Amazon CloudWatch – Monitoring

CloudWatch tracks system performance and logs. It helps in alerts, debugging, and optimization.

Amazon Athena – Querying Data

Athena lets you run SQL queries on S3 data. No servers are needed—pay only for queries run.

AWS IoT Core – Connected Devices

IoT Core connects devices to the cloud securely. It allows devices to send, receive, and process data.

Amazon SageMaker – Machine Learning

SageMaker helps build, train, and deploy ML models. It supports notebooks, training jobs, and real-time inference.